

Sound

Sounds are made when something vibrates, changing movement energy into sound energy. Sound travels through solids, liquids, and gases as waves. The waves are caused by the vibrations, which make particles knock into each other. Sound cannot travel in space because there are no air particles for it to travel through.

Sound waves

Sound travels in waves that first squash particles together, then spread them out. It's a ripple of energy that spreads out from its source. When a wolf howls, folds inside its throat vibrate, sending sound waves through the air.

The particles spread out after being squashed.

The vibrations squash particles together.



Decibels

We measure the energy of a sound in decibels. Here are some familiar sounds with their volume in decibels.

Human whisper
across room



Acoustic guitar
1 m (3 ft) away



Car 1 m
(3 ft) away



By speakers
at a concert



10 20 30 40 50 60 70 80 90 100 110